



Explore 145 years of NASA temperature data revealing accelerating climate change. See the clear correlation between human CO2 emissions and global warming.

**+1.47°C**

Current Global Warming  
Since pre-industrial times

**+1.28°C**

NASA GISS Temperature Anomaly  
Since 1951-1980 average baseline

**+0.11°C**

Warming Rate/Decade  
Accelerating since 1980

**2024**

Warmest Year on Record  
Last 10 years were warmest ever recorded

**424**

Current CO2 (ppm)  
Highest in 3+ million years

 Data Controls

Time Range

 All Time (1880-2024)

 Recent Era (2000-2024)

 Last Decade (2014-2024)

Analysis Type

 Temperature Trend

 CO2 Correlation

Current Selection:

145 years of data  
Showing temperature trend

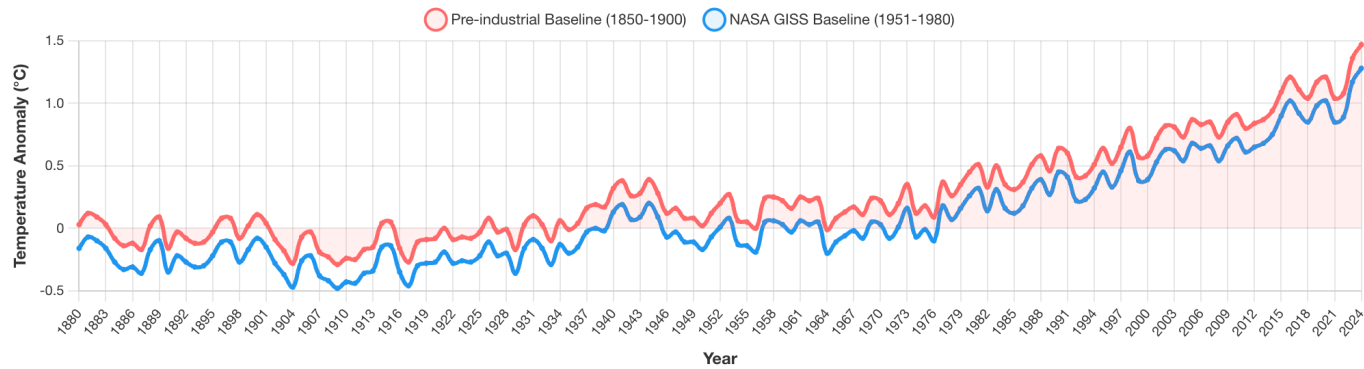
**NASA GISTEMP Global Temperature Anomalies**

**Data Source:** NASA Goddard Institute for Space Studies (GISTEMP v4)

**Baseline:** Pre-industrial (1850-1900) average temperature

**Time Range:** 1880-2024 (145 years)

 Sustainability Chatbot



### Key Finding

Temperature rise has accelerated dramatically since 1980, with warming rate increasing from 0.11°C/decade (1880-2024 average) to 0.28°C/decade (1980-2024). The last 10 years show unprecedented warming, with 2024 reaching 1.47°C above pre-industrial levels - the hottest year on record.

### About This Project

Created for the **Viz Con 2025** contest at **Analyticon**(Amazon-wide Conference on Data Engineering, Science and Analytics). This interactive application explores the connections between human activity and environmental conservation through compelling data narratives using real datasets.

### Data Sources



#### Temperature Data:

NASA GISTEMP v4 (Goddard Institute)



#### Renewable Energy:

IRENA Global Energy Statistics



#### Critical Minerals:

IEA Critical Minerals Data Explorer



#### Sea Ice Data:

NSIDC Sea Ice Index



#### CO2 Data:

NOAA Mauna Loa Observatory

### Technology & AI

**Frontend:** React.js, D3.js, Chart.js, Framer Motion

**Backend:** Python Flask, NumPy, Pandas

**AI Integration:** Pattern recognition, correlation analysis, predictive modeling, and natural language insight generation

**AI Chatbot:** Amazon Bedrock (Nova Pro Model), API Gateway, AWS Lambda, S3, IAM

**Deployment:** AWS Amplify with automated CI/CD pipeline

### Viz Con 2025 Entry

Entry for the **2025 Viz Con contest**, part of the **Analyticon** - Amazon-wide Conference on Data Engineering, Science and Analytics

**Challenge:** "Visualizing a Sustainable Tomorrow"

**Approach:** Multi-dataset integration with real NASA GISTEMP, NOAA, IRENA, and IEA data

**Goal:** Inspire climate action through compelling data storytelling

**Features:** Interactive visualizations, correlation analysis, predictive modeling, and accessibility-first design

Built with ❤️ by @ygal and his best friend Amazon Q 🤖

Powered by AWS | NASA, IRENA, IEA, NSIDC & NOAA data